APhO 2023, Problem T1 (Answer Sheet for Part E)

In your calculations, use the value $n_{\text{ion}} = 10^{13}\,\text{m}^{-3}$ for the number density of the oxygen ions in the upper atmosphere (instead of the value $n_{\text{ion}} = 10^{12}\,\text{m}^{-3}$ given in the problem statement).

Table 1

h [km]	T_h^{air} [day]	u_h^{air} [m/day]	u_h^{ion} [m/day]	u_h^{ind} [m/day]	u_h^{total} [m/day]	w_{ISS} [m/day]
350						145
375						N/A
400						N/A
410						70

Here, u_h^{total} is defined as the sum of u_h^{air}, u_h^{ion}, and u_h^{ind}. The quantity w_{ISS} is the observed rate of descent at a given altitude, as extracted from the blue excerpts in the problem statement. The values of w_{ISS} are provided only for reference.

Table 2

h [km]	H_h^{air} [m]	H_h^{ion} [m]	H_h^{ind} [m]
350			
375			
400			
410			

APhO 2023, Problem T1 (Answers for Part E)

Table 1

h [km]	T_h^{air} [day]	u_h^{air} [m/day]	u_h^{ion} [m/day]	u_h^{ind} [m/day]	u_h^{total} [m/day]	w_{ISS} [m/day]
350	330	170	6.6	26	203	145
375	2400	29	6.6	26	62	N/A
400	19000	5.0	6.6	26	38	N/A
410	42000	2.5	6.6	26	35	70

We have used $\mu_{\text{O}} = 0.016\,\text{kg/mol}$ for the molar mass of the oxygen ions. The numerical results for T_h^{air} are very sensitive to the choice of approximations when evaluating the relevant integral. Thus, even large deviations from the T_h^{air}-values listed here are acceptable.

Table 2

h [km]	H_h^{air} [m]	H_h^{ion} [m]	H_h^{ind} [m]
350	11	0.4	1.6
375	1.9	0.4	1.7
400	0.3	0.4	1.7
410	0.2	0.4	1.7

We see that for $h > 380\,\text{km}$, the dominant effect is the drag due to the Earth's magnetic field. Much weaker is the drag due to collisions with oxygen ions. Finally, the weakest effect is the drag due to collisions with neutral molecules.

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